



# Alcohols

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# Alcohols (R-OH)

In alcohol, a hydroxyl group (-OH) replaces a hydrogen atom in an alkane. The ending (-e) is changed to (-ol).



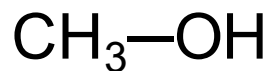
3-Buten-1-ol

# Naming Alcohols

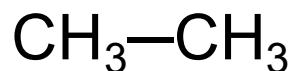
- **IUPAC name:** We replace the **-e** in alkane name with **-ol**.
- **Common name:** As simple alcohols, using the name of the alkyl group followed by “*alcohol*”.



methane



methanol      (methyl alcohol)

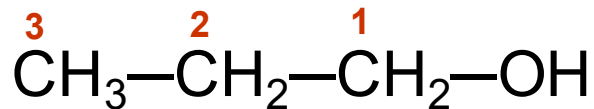


ethane

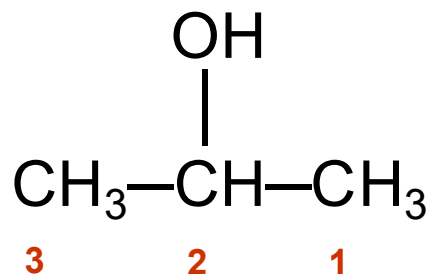


ethanol      (ethyl alcohol)

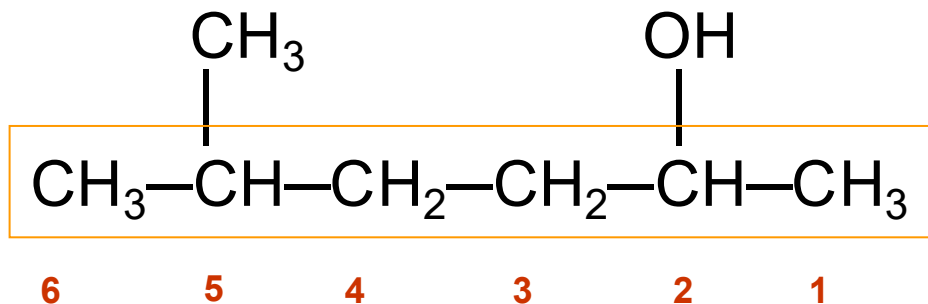
# Naming Alcohols



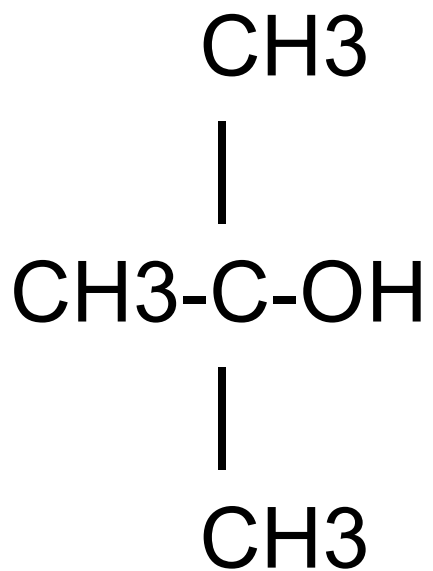
1-propanol



2-propanol (isopropanol)



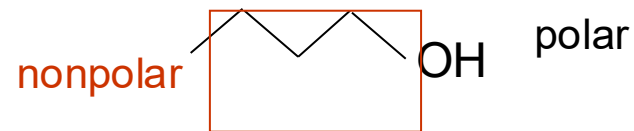
5-methyl-2-hexanol



2-methyl-2-propanol

# Physical Properties of Alcohols

1. Hydrogen bonding between alcohol molecules.
2. Have higher boiling points than Alkanes, Alkenes, and Alkynes.
3. More soluble in water (Molecular weight  $\uparrow$  : solubility  $\downarrow$ ).
4. Alcohols are polar molecules.
5. The carbon side of alcohols is nonpolar, while the OH side is polar.

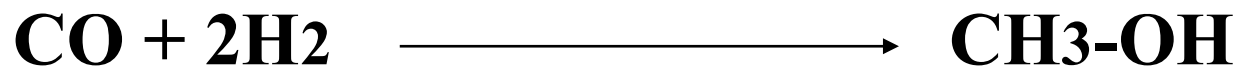


# Some important alcohols and their uses:

## 1-Methanol (CH<sub>3</sub>OH)

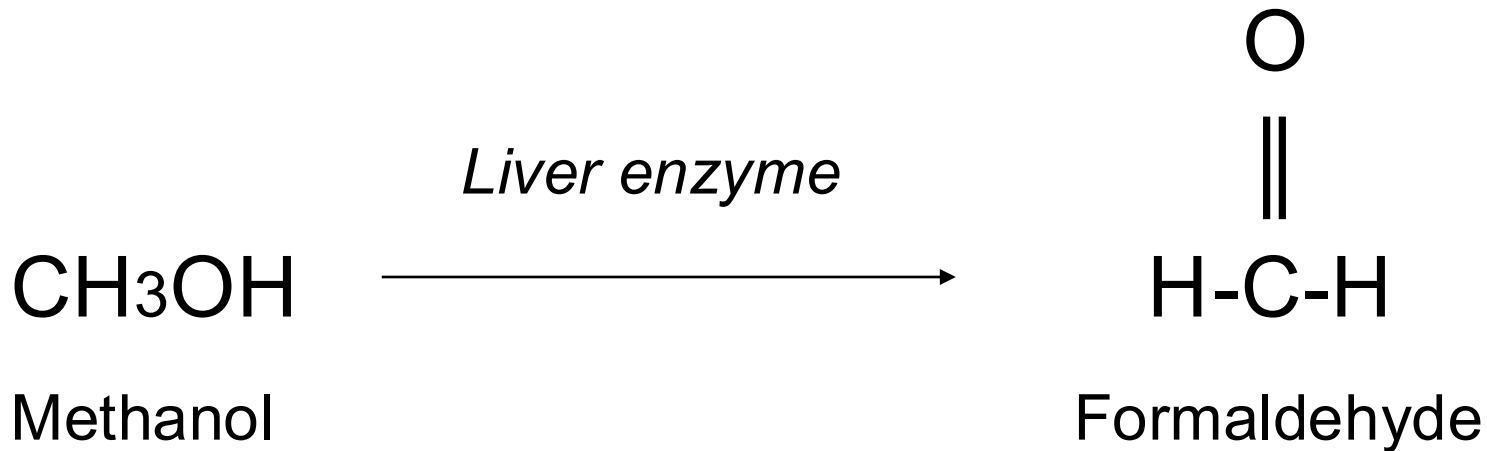
- Is the simplest alcohol found in many solvents and paint removers.
- It is sometimes called wood alcohol because Methanol is produced when wood is heated at high temperatures in the absence of air.
- Methanol is prepared commercially by reacting carbon monoxide with hydrogen gas at high temperatures and pressures.

200-300 °C / 200Atm



**ZnO/Cr<sub>2</sub>O<sub>3</sub> Catalyst**

**Methanol** is a poisonous liquid; if ingested, it is oxidized to formaldehyde, which causes headache, blindness, and death, because react rapidly with proteins of cells and cause denaturizing.

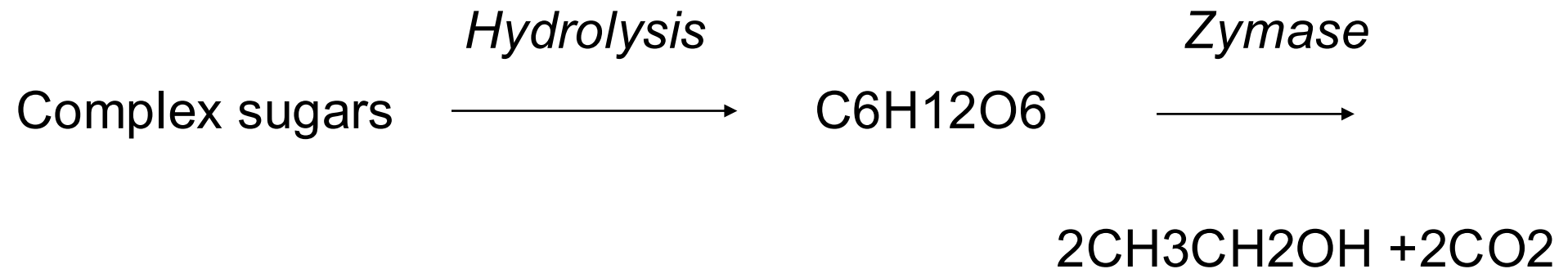


## 2-Ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ )

- In the hospital, the word alcohol means ethanol, one important property of ethanol is its ability to denature proteins. Because of this property, ethanol (70%) is widely used as an antiseptic.

- Ethanol is widely found in most homes as a solvent for **pharmaceuticals** such as, (Tinctures iodine in alcohol), colognes, perfumes, mouth washes, and hair sprays. ‘Gasohol’ is a mixture of ethanol and gasoline used as a **fuel**.

- Ethanol is used as a beverage; most alcoholic drinks are made by fermentation of starch or sugars.



- Ethanol is toxic; extensive ingestion over along period of time leads to:

1- Cirrhosis.

2- Loss of memory.

3- Strong physiological addiction.

### 3- 2-propanol (isopropanol)

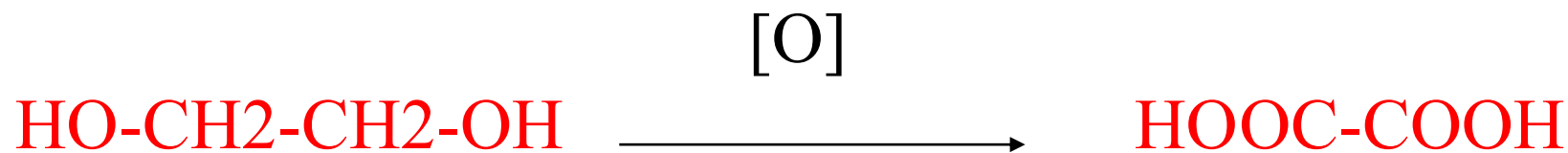


- Used as an astringent antiseptic, because it evaporates rapidly and cools the skin, reducing the size of blood vessels near the surface.
- Used to clean the skin before injection or taking a blood sample. It is also used to sterilize equipment because it destroys bacteria by coagulating protein.

#### **4- Ethylene glycol (ethanediol) (dihydroxy alcohol) $\text{HO-CH}_2\text{-CH}_2\text{-OH}$**

- Is used as antifreeze in heating and cooling systems, it is also a solvent for paints, inks, and plastics, and is used in the preparation of fibers such as Dacron.

- It is extremely toxic if ingested in the body, as it is oxidized to **oxalic acid**, which forms insoluble salts in the kidneys that cause renal damage, convulsions, and death. Because its sweet taste is attractive to pets and children, ethylene glycol solutions must be carefully stored.



# Fomepizole (4-methylpyrazole)

- Is a medication used to treat methanol and ethylene glycol poisoning. It may be used alone or together with hemodialysis. It is given by injection into a vein.
- Common side effects include headache, nausea, sleepiness, and unsteadiness.
- Fomepizole works by blocking the enzyme that converts methanol and formaldehyde to their toxic breakdown products.

# 5-Glycerol, Glycerin, (1, 2, 3-propanetriol)

## [tri hydroxyl alcohol]

- Is a constituent of fats and is produced from the hydrolysis of all fats.



- It is used in the production of nitroglycerin (Glyceryl trinitrate).

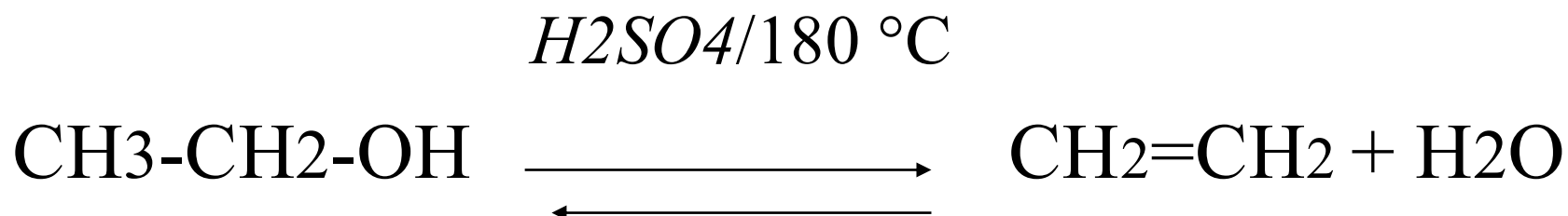


# Nitroglycerin

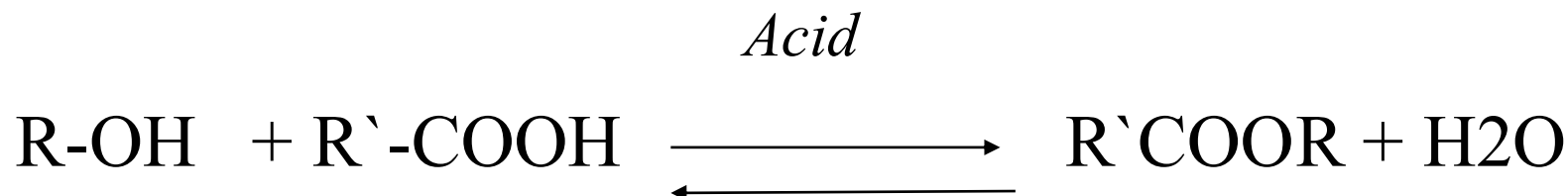
- Is used to relieve angina pectoris (pain in the chest). It relaxes cardiac muscles and smooth muscles in the smaller vessels; it is provided in small tablets under the tongue or applied to the skin as a cream.

# Reactions of alcohols:

## 1-Dehydration



## 2-Ester formation

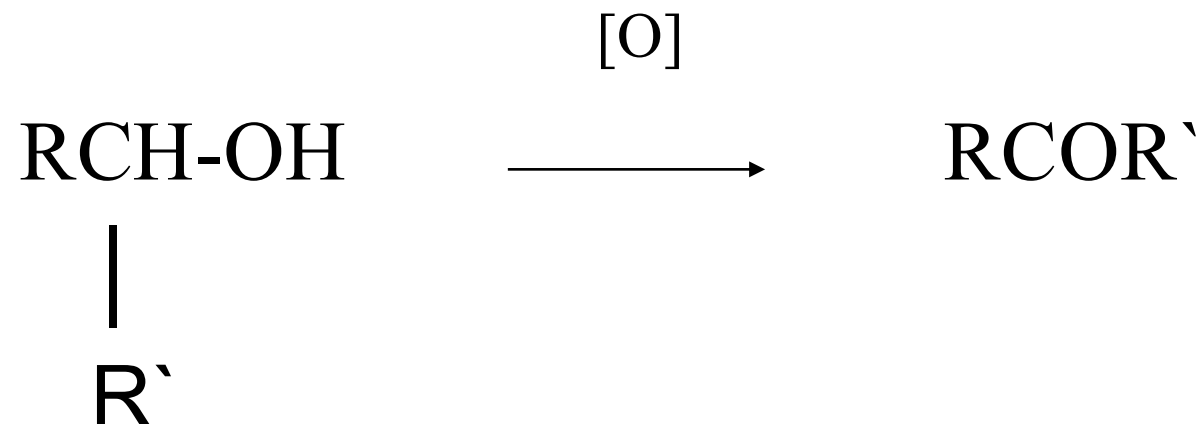


# 3-Oxidation

Primary alcohols are oxidized to aldehydes



Secondary alcohols are oxidized to ketones



# Oxidation of alcohols in living systems

- An example is the oxidation of malate to oxaloacetate.

*Malate dehydrogenase*

